

Madhusudhan Reddy Pittu

Curriculum Vitae

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📁 mathrulestheworld.github.io

Summary

- Postdoctoral Fellow at Georgia Tech ISyE, working with Mohit Singh on combinatorial optimization and approximation algorithms.
- Ph.D. in Computer Science from CMU; publications at STOC, FOCS, SODA, and ICALP. Research in clustering, experimental design, constrained subspace approximation, and comparison-based optimization.
- Current AI research studies unintended behavioral changes from LLM fine-tuning through controlled experiments, gradient analysis, and parameter interventions.
- Hands-on PyTorch and Hugging Face experience with fine-tuning, learned prompts, and behavioral evaluation. Interested in reliable AI and impactful real-world applications.
- Bronze Medalist representing India at the 57th International Mathematical Olympiad (IMO).

Skills

Theory	Algorithm Design and Analysis, Combinatorial Optimization, Integer and Linear programming, Convex Optimization, Linear algebra, Probability and Statistics, ML theory.
ML / AI	PyTorch, Hugging Face Transformers, full-parameter and parameter-efficient fine-tuning (soft prompts, prefix tuning, LoRA), behavioral evaluation, gradient analysis, SLURM/HPC
Languages	Python, C++

Education and Experience

Sept.–Dec. 2026	Postdoctoral Fellow , <i>Georgia Institute of Technology</i> , Atlanta. H. Milton Stewart School of Industrial and Systems Engineering (ISyE) Host: Prof. Mohit Singh
2025–2026	Postdoctoral Associate , <i>Courant Institute, New York University</i> , New York. Host: Prof. Anupam Gupta
2021–2025	Ph.D. in Computer Science , <i>Carnegie Mellon University</i> , Pittsburgh. Advisors: Prof. Anupam Gupta and Prof. David P. Woodruff
2017–2021	B.Tech. in Computer Science and Engineering , <i>Indian Institute of Technology Kharagpur</i> , India. Advisor: Asst. Prof. Palash Dey

Recent Projects

2026–present	LLM Fine-Tuning, Generalization, and Alignment. <i>Ongoing, unpublished research (Weird Generalization project).</i> Study how narrow fine-tuning changes held-out behavior in Qwen models. Use controlled full-parameter/LoRA experiments, gradient analysis, and parameter transplants; evaluate baseline-behavior regularization across paired seeds while measuring intended task learning and unwanted changes separately.
Spring 2026	Prompt Boosting for LLM Reasoning. Investigated learned soft and hard prompts for LLM reasoning. Implemented fine-tuning and evaluation pipelines for Qwen and Gemma models.

Industry Visits & Internships

- Spring 2025 **Student Researcher**, *Google*.
Hosts: Sreenivas Gollapudi, Kostas Kollias
- Summer 2023 **Non-Degree Visiting Student**, *Toyota Technological Institute at Chicago*.
Hosts: Ali Vakilian, Yury Makarychev, Julia Chuzhoy
- Fall 2022 **Visiting Graduate Student**, *Simons Institute for the Theory of Computing*.
Data-Driven Decision Processes program
- Summer 2020 **Visiting Student Research Internship (Remotely)**, *Georgia Tech*, Atlanta, USA.
Hosts: Prasad Tetali and Mohit Singh
- Summer 2019 **Narendra Internship**, *Indian Institute of Science*, Bengaluru.
Host: Arindam Khan

Selected Publications

Full publication list on page 3 and at mathrulestheworld.github.io.

- STOC '26 Combinatorial Optimization using Comparison Oracles
with Vincent Cohen-Addad, Tommaso d'Orsi, Anupam Gupta, Guru Guruganesh, Euiwoong Lee, Renato Paes Leme, Debmalya Panigrahi, Jon Schneider, and David P. Woodruff
[Optimization from Human Feedback](#)
- ICALP '25 Guessing Efficiently for Constrained Subspace Approximation
with Aditya Bhaskara, Sepideh Mahabadi, Ali Vakilian, and David P. Woodruff
[Constrained PCA / Dimensionality Reduction](#)
- SODA '24 Approximation Algorithms for Weighted Nash Social Welfare via Convex and Non-Convex Programs
with Adam Brown, Aditi Laddha, and Mohit Singh
[Fair Resource Allocation](#)
- FOCS '23 The Price of Explainability for Clustering
with Anupam Gupta, Ola Svensson, and Rachel Yuan
[Explainable Clustering](#)
- FOCS '22 Determinant Maximization via Matroid Intersection Algorithms
with Adam Brown, Aditi Laddha, Mohit Singh, and Prasad Tetali
[Experimental Design / Subset Selection](#)
- SODA '22 A 3-Approximation Algorithm for Maximum Independent Set of Rectangles
with Waldo Gálvez, Arindam Khan, Mathieu Mari, Tobias Mömke, and Andreas Wiese
[Computational Geometry](#)

Awards and Achievements

- Secured **Bronze Medal** representing India at the **57th International Mathematical Olympiad (IMO)**, 2016
- **IITKGP Foundation-USA International Internship Award**, 2020
- **Infosys Award** for excellent performance in International Olympiads, 2016
- **Best Solution Award** in IMO training camp, 2015

Complete Publication Record

Peer-Reviewed Journal Articles

- 2026 **A $(2+\varepsilon)$ -Approximation Algorithm for Maximum Independent Set of Rectangles.**
with Waldo Gálvez, Arindam Khan, Mathieu Mari, Tobias Mömke, and Andreas Wiese
ACM Transactions on Algorithms, online publication, July 2, 2026.
- 2025 **Approximation Algorithms for the Weighted Nash Social Welfare via Convex and Non-Convex Programs.**
with Adam Brown, Aditi Laddha, and Mohit Singh
Mathematical Programming, Series A, online-first, December 23, 2025. Journal version of the SODA 2024 paper.

Peer-Reviewed Conference Papers

- 2026 **Combinatorial Optimization using Comparison Oracles.**
with Vincent Cohen-Addad, Tommaso d’Orsi, Anupam Gupta, Guru Guruganesh, Euiwoong Lee, Renato Paes Leme, Debmalya Panigrahi, Jon Schneider, and David P. Woodruff
58th ACM Symposium on Theory of Computing (**STOC**), pp. 1995–2006.
- 2025 **Max-Cut with Multiple Cardinality Constraints.**
with Yury Makarychev and Ali Vakilian
APPROX/RANDOM, LIPIcs 353, Article 13, pp. 13:1–13:21.
- 2025 **Guessing Efficiently for Constrained Subspace Approximation.**
with Aditya Bhaskara, Sepideh Mahabadi, Ali Vakilian, and David P. Woodruff
52nd International Colloquium on Automata, Languages, and Programming (**ICALP**), LIPIcs 334, Article 29, pp. 29:1–29:20.
- 2024 **Approximation Algorithms for the Weighted Nash Social Welfare via Convex and Non-Convex Programs.**
with Adam Brown, Aditi Laddha, and Mohit Singh
ACM-SIAM Symposium on Discrete Algorithms (**SODA**), pp. 1307–1327.
- 2023 **The Price of Explainability for Clustering.**
with Anupam Gupta, Ola Svensson, and Rachel Yuan
IEEE Symposium on Foundations of Computer Science (**FOCS**), pp. 1131–1148.
- 2022 **Determinant Maximization via Matroid Intersection Algorithms.**
with Adam Brown, Aditi Laddha, Mohit Singh, and Prasad Tetali
IEEE Symposium on Foundations of Computer Science (**FOCS**), pp. 255–266.
- 2022 **A 3-Approximation Algorithm for Maximum Independent Set of Rectangles.**
with Waldo Gálvez, Arindam Khan, Mathieu Mari, Tobias Mömke, and Andreas Wiese
ACM-SIAM Symposium on Discrete Algorithms (**SODA**), pp. 894–905.
- 2020 **On Guillotine Separability of Squares and Rectangles.**
with Arindam Khan
APPROX/RANDOM, LIPIcs 176, Article 47, pp. 47:1–47:22.

Additional Public Preprints

- 2025 **An Algorithmic Upper Bound for Permanents via a Permanent Schur Inequality.**
with Aditi Laddha
Preprint, arXiv:2509.08121, 2025.
- 2022 **Efficient Determinant Maximization for All Matroids.**
with Adam Brown, Aditi Laddha, and Mohit Singh
Preprint, arXiv:2211.10507, 2022.

Selected Talks and Presentations

- 2026 Combinatorial Optimization using Comparison Oracles, STOC, June 24, 2026.
- 2026 Geometric Methods for Combinatorial Optimization Using Comparison Oracles, NYU Geometry Seminar, March 10, 2026 (video).
- 2025 Combinatorial Optimization using Comparison Oracles, Yale Theory Student Seminar, October 24, 2025.
- 2025 Max-Cut with Multiple Cardinality Constraints, APPROX, August 12, 2025.
- 2025 Guessing Efficiently for Constrained Subspace Approximation, ICALP, July 10, 2025.
- 2023 The Price of Explainability for Clustering, FOCS, November 7, 2023; also presented at CMU Theory Lunch, October 25, 2023.
- 2022 Efficient Determinant Maximization for All Matroids, IISc-MSR/Bangalore Theory Seminar, Indian Institute of Science, December 23, 2022 (video).
- 2022 A 3-Approximation Algorithm for Maximum Independent Set of Rectangles, SODA 2022.
- 2020 On Guillotine Separability of Squares and Rectangles, APPROX/RANDOM, August 17, 2020 (video).